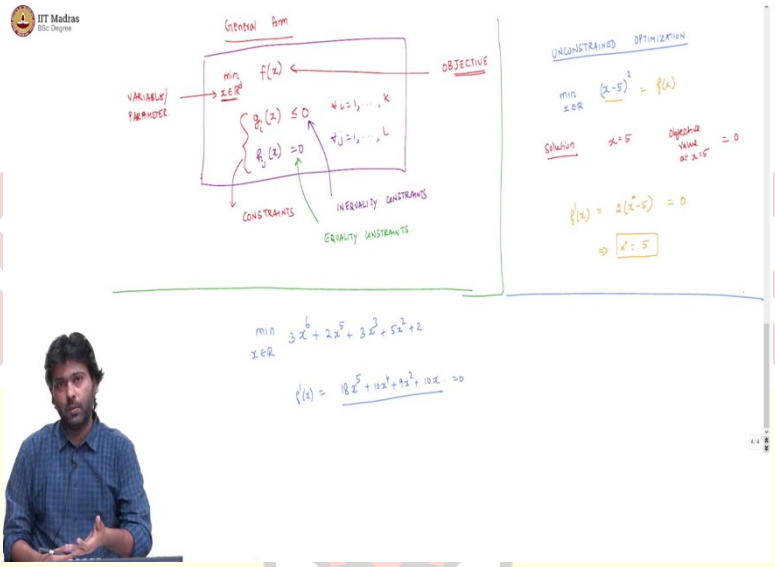


IIT Madras

ONLINE DEGREE

Machine Learning Foundations
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Solving an unconstrained optimization problem (Part 1)

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So, now we need to understand how to solve an optimization problem that we have set up already. So, the general form of the optimization problem has an objective and a bunch of constraints. That is the most general form. Maybe it makes sense to start with a form where you do not have any constraints.

Take a very, very simple example. Try to understand how to solve such a problem. And then maybe we will slowly move on to adding constraints and understand how one can solve a problem if you have constraints as well. This is, this would be our line of approach to understand this problem.

To begin with, we are going to start with a very extremely simple problem for which all of you know the answer to, but still we will want to understand it from an optimization lens and it will reveal a lot of interesting ideas. The problem that we will start considering is what we will call this an unconstrained optimization problem. Where by unconstrained I mean that you do not have any constraints. So, this gi of x and hj of x are not there in the picture.

Let us take the simplest possible example that we can think about. So, let us start with the problem, you want to minimize over x in R, which means that it is not even a two dimensional variable, like in the cow example, it is a very simple real number that you want to find where

the function of interest is, well, x - 5 square. Well, what is the x that solves this problem, and what is the value of the function at that point. It should be obvious by looking at this. If not just pause and think about this.

But it should be very clear that solution to this problem is x equals 5 and objective value at that point x equal to 5 is 0. And of course, because we are minimizing and this is a squared function, it has to be necessarily non-negative. It cannot be negative values. And we have found a point x equals 5 for which the value is 0, which means that it necessarily has to be the minimum. So, no other point can achieve something less than 0 for this problem. And hence, this is optimum.

Of course, this is a different line of reasoning. We cannot really hope to ask a computer to potentially do this way of reasoning things. But if you had to solve this problem, in the traditional way that we have been taught at school, how would we go about solving this problem?

Well, the simplest way we will do is that, well, we will think of this as f of x and we would take the first derivative of f of x, in this case, f dash of x, if this is f of x. You will take f dash of x, compute f dash of x, which should be like 2 × x - 5, in this case, set f dash of x to 0. Well, in this case, set this to 0, which means that for that particular x star where this becomes 0 is the optimal value and that is just 5. This is a perfectly reasonable way to solve this problem.

Just that this way of solving problems does not generalize to any, all problems that you can potentially be interested in. So, in this particular case, it was easy. You took the first derivative, set it to 0 and you were able to solve the equation that came out. There are many, many problems and we will see a lot more complicated examples when we tie up this with machine learning. But for the moment, I will just give you a simple example where it might be hard to solve this problem by using this particular method. So, and here is a problem.

So, let us say I want to minimize x in R again. And now the question that I am interested in is, let us say 3x power 6 + 2x power 5 + 3x cube + 5x square + 2. I want to let us say solve, find the minimum of this function. This is my objective. How do I do this? Well, if you went through the school method, then you are going to take derivatives which means that if I do f dash of x, that is going to be. well, in this case, 18x to the 5 + 10x to the 4 + 9x square + 10x.

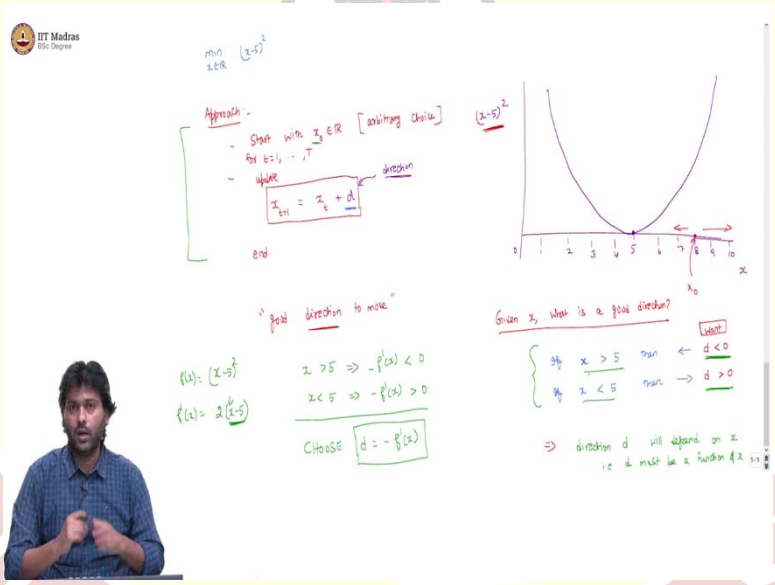
Now, I will set f dash of x to 0 and that will want me to solve an equation in degree 5. And I do not know of any easy way to do that. So, I need to solve this equals 0. So, what value of x does this function becomes 0? I do not really know. So, it is not clear how to solve this. It is

not even clear if whatever I get is, in fact, the minimum and so on. So, this is not a method that will always work.

So, we need a more general method and this general method should also be easy for, let us say, a computer to implement. So, that is the goal. It is not that our reasoning said that, well, this function is always positive, for instance, x - 5 square and so the value of x has to be, that minimizes this as to be 5 is that is a logical reasoning that we did, which you cannot teach computer do directly.

So, you need some other systematic procedure that you want the computer to follow, such that it will get to the minimum of this particular function and that is what we are going to see next. How do you come up with a systematic procedure so that the computer will be able to solve this problem?

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So, how do we make a computer solve a problem, which we just looked at which is minimize over x in R, x - 5 square. So, here is a potential way to do this. This is an iterative way that we are going to think about. And the approach that we are going to take is very simple. And it is going to look like this. Start with some x0. So, let me make the notation slightly clearer, x0 in R. Now, you have the function you can understand anything about that function at x0 equal to R and this x0 that we are picking at this point, do not worry about how we are taking this x naught, let us say can be an arbitrary choice.

Let us say you pick an arbitrary x0 maybe, let us say, 10. So, it can be any arbitrary value. And now we want to somehow improve this value that we have currently. So, we want to move this

x naught or x0 to x1, the next point that we are going to consider such that the next point, hopefully, is a better point or a better guess for the minimum of this problem.

So, now, we want to update x(t + 1), let us say this is an iterative algorithm for t equals 1 to dot, dot, dot, capital T, whatever number of rounds, let us say you want to run this for 1000 rounds, we will talk more about this capital T a little bit later. But for the moment, let us say we fix 1000 rounds, and then we are going to update our current guess to a new guess based on something.

So, well, what should we do? We have a current guess, which is xt, and we want to somehow make this guess better. So, one way to make this guess better is to add something to this which we were going to call as d to indicate a direction and we will talk a bit about this such that we hope that if you are moving in the right direction, then your next point is a better point than your current point. That is the hope. And you keep doing this. As you get better and better then eventually the hope is that you will reach the minimum of this function.

So, this is a very gentle approach. So, we just relies on a single equation, which is you are in a particular position currently in the parameter or the variable space and you want to move to a different position, which means that you need to find a good direction to move. So, you want a good direction. Now, of course, what is a good direction, a direction which will hopefully take us close to the minimum. So, now, I am still being vague with respect to what does it mean to say something is a direction.

Let us take for this example, particular example and then we will come to other examples a little bit later, this particular simple example. If I had to plot this graph of x - 5 square, how do you think it will look like. If you have not seen this before, you should pause and try to plot this. So, here is 0, here is 1, 2, 3, 4, 5, 6, and so on.

So, maybe I will add a few more. Now, how is this graph going to look like? Well, this graph is going to be, if you have seen parabolas, this is going to be a parabola which will, sorry, about that. This is my x, this is my x - 5 squared and this is, this point is my 5, where we (()) (10:40) just the value of 0.

Of course, you can plot this and see that, hey, the value is 5, which the value at 5 is where it is minimum, but that is not the goal. So, the goal is more to say, how do I automatically find this value as I improve my x. So, we started with some x. Let us say we are at this particular x,

which is x naught. Let us say we are at some x, which is let us say x naught. And now we want to understand, we want to find a good direction to move.

So, now, what do I mean by direction? In this particular picture what is a direction? So, you are at x naught. There are how many directions are there for you to move from x naught? Think about it. Well, if you think about it, there are not too many directions. So, you can either move from x naught to the left or you can move from x naught to the right.

What does it mean to say you are moving to the left? Which means that in this case, you are adding a negative value to x naught. That is what it means to say that you are moving to the left. So, if I am at 8, if I add - 3, then I moved to the left by 3 units, which means that I have reached 5. On the other hand, if I add 2, then I move to the right by 2 units, which means that I am, I have reached 10. So, there are only two directions, and these directions correspond to adding either a positive quantity or a negative quantity to my current point.

Now, for this particular problem, let us ask the question, what is a good direction to move given a point x naught or in general x. So, the question is given x, what is a good direction? We are still in this particular function x - 5 square. Let us talk about this function and then generalize it to other functions in a bit. What is a good direction?

Well, for this particular function, it is pretty clear what is a good direction. So, here is the rule so to say. So, if x whatever your current direction is greater than 5, sorry, your current point is greater than 5, then you know moving to the left is a good thing to do, because you are getting closer to your minimum, which is 5. If x is less than 5, of course, moving to the right is a good thing to do.

Well, we know that moving to the left means decreasing your value, which means that adding a negative value, so that is adding some negative value to your d is negative. And if moving to the right means your d is positive, which means that I can equivalently say, if x is greater than 5, then I need d to be less than 0; want.

So, let me emphasize that we want a d such that d is less than 0. If x is less than 5, then we want a d such that d is greater than 0. Of course, nobody is giving us this d. So, we need to understand how to get such a d. That is the whole game in optimization. So, you are at a point you need to understand a good direction to move. Now, we have identified at least for this problem, if x is greater than 5, then you want to add a d such that d is less than 0. And if x is less than 5, you want to add a d such that d is greater than 0.

The first thing that comes out of this is the understanding that the direction d will depend on x. This is an obvious statement once you think about that. So, depending on where you are, the direction you want to move is going to be different. So, whatever way we are picking directions, it cannot be a constant direction always. Depending on which point we are in, our direction has to be different.

So, that is what this is saying, because for x is greater than 5, we want a value which is less than 0, for x is less than 5, we want a value which is greater than 0, which means that d is also a function of x. That is what this means. So, direction of d depends on x simply means that d is a function of x, that is d is a function of x or rather, I should say, d must be a function of x.

So, we still do not know what function of x this is actually. But we want to come up with a function which will work, at least for this particular example to begin with. Now, how to choose the d. So, how do you choose such a direction for this example? Now, here is a place where you can pause and think, can you come up with functions such that that function will take a negative value of x is greater than 5 and a positive value if x is less than 5.

Well, if you thought about this, you can come up with innumerable number of functions which will do the trick. Now, what I am going to do now, and it might feel that I am pulling this function, I mean, pulling this off the bag just like that, but we will justify the choice of this function as we go along. So, what I want to do is, I am going to look at the derivative of the original function at the point of interest.

So, let us say we are at a particular x. The function, remember, is x - 5 squared. This is f of x. So, which means that my f dash of x is going to be 2 × x - 5. So, let us say we are at an x which is 8. So, or, so let us take 8 as an example. Now, what is my f dash of x going to be? Well, f dash of x is going to be 2 × 8 - 5, so which is 2 × 3 which is 6 in this case. Well, in general, if you look at this, why did this 6 come about, is 6 is a positive number. And why is it positive, because 8 is greater than 5.

So, the only thing that determines the sign of this particular function is whether 8 - 5, so wherever I have x, whether x - 5 is positive or not, which means that in this particular function for this specific case of x - 5 square, it so turns out that f dash of x will always be greater than 0 when x is greater than 5 and will always be less than 0 when x is less than 5. So, when x is greater than 5, this is going to be a positive quantity and so 2 × that will also be a positive quantity.

Similarly, if x is less than 5, well, then 2 × x - 5 is going to be a negative quantity, which is, which does not quite do the job for us. What we want is the opposite of this. If x is greater than 5, what we have found now is if x is greater than 5, f dash of x is also greater than is greater than 0. And if x is less than 5, f dash of x is less than 0.

On the other hand, what we really want is that if x is greater than 5, we want d to be less than 0. And if x is less than 5, we want d to be greater than 0. We want the opposite way to happen. Well, how do you make this opposite thing to happen? Pause and think for a second, you will be able to get the answer.

Well, one way to think about this is, I am going to choose and this is the big point we will start with, we are going to choose d as - f dash of x. We know d must be a function of x. And the specific function that I am going to choose for the moment is d is - f dash of x. So, what property does d equals - f dash of x how?

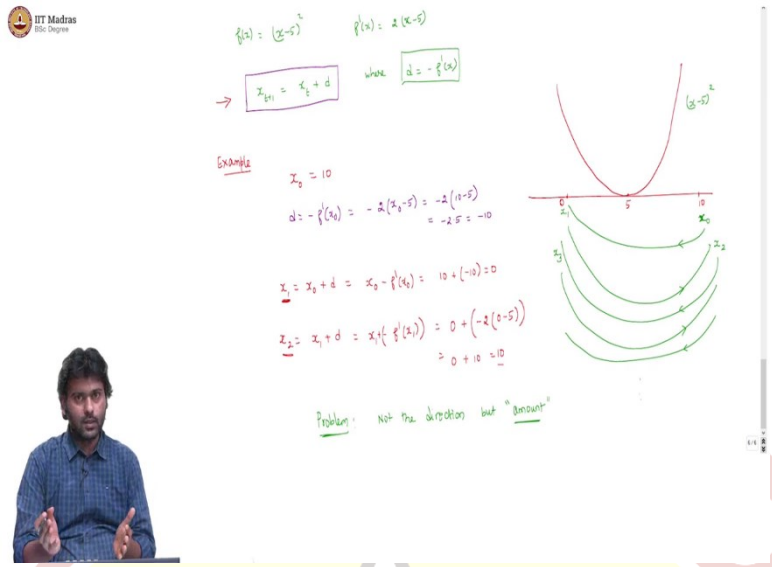
Well, if x is greater than 5, then - f dash of x is going to be less than 0, and - f dash of x will going to be greater than 0 if x is less than 5, which means that we have managed to find a function no matter what x you give, you will essentially move in the correct direction, which is what we wanted by looking at the picture here at least.

So, which means what is going to be our algorithm at this point? Well, we are going to start from some arbitrary x naught and then we are going to find the derivative of the function at x naught, take its negative and add it to x and keep doing this iteratively. So, this is our current algorithmic procedure and a computer can do this, because all you want to do is to be able to compute derivatives.

If you can write a function that can compute derivatives of a function, then you can keep calling that at different points and then iteratively keep updating your x. So, it is a simple thing to write this as a function as long as you can compute derivatives easily. But does this do the job? Well, there are several questions that you should be asking at this point.

One is, well, it may have worked for this function, does it work always? Well, even for this function what is the guarantee that we are going to get to the minimum? I mean, are we sure that this update works and so on. We are going to talk more about how good is this particular direction that we have picked so far by looking at an illustrative example, which will throw light × whether our current algorithm is a good algorithm or not.

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So, let us go forward and understand what we are trying to do. So, we have a function f of x, which is x - 5 squared and we know f dash of x is 2 × x - 5 and our update is, just to be consistent with notation, xt + 1 is xt + d, where we are saying the direction that we are going to move is negative of f dash of x. This is where we are at this point. Let us see if this actually works. So, let us try an example and see if this is a good update rule or are we missing something from this. Here is an example.

As I mentioned, we can start with any arbitrary x naught. There is no restriction on it. Let us say we start with x naught. Again, let me not use capitals, let me in small letters. Let us say x naught is 10. And what I am going to do now is slowly go over this procedure and see what happens. So, in parallel, let us also look at the graph. So, this is 5, let us say, we are starting from. This is our x naught at this point. So, this is where we are. Now, let us try to compute the derivative, let us try to compute the direction at this point.

Well, d is - f dash of x naught, which is - 2 × x naught - 5, which is - 2 × 10 - 5, which is - 2 × 5, which is - 10. So, this is my d, which means that my x1 using this rule, update rule that I came up with, is x naught + d, which is equal to x naught - f dash of x naught, which in this case is 10 -, well, - f dash of x naught happens to be - 10, which means 10 + - 10 which is 0.

So, my next point that I going to look at happens to be 0. So, which means that we are at x naught here and we have moved to the point 0 as our next point of interest. So, this becomes my x1, which is good. So, we have made a move from x naught to x1. Now, we are going to continue and then try to understand what is going to be x2. So, x2, how do you get x2? Well, you start with x1 and then you add the d corresponding to x1. So, it is not the same d.

We know that d is a function of x itself. Well, the d should correspond to x1, which is x1 - f dash +, well, - f dash of x1, which is, well, we are at 0 +, we want to look at the - f dash of x1. Well, what is - f dash of x1? It is going to be - 2 × x naught, x1 - 5 which is 0 - 5. So, this is going to be 0 + - 2 × - 5, which is 10 and this is going to be 10.

So, my x2 is going to be, I start with 0, I added 10, so my x2 is going to be 10. So, in the picture I am here. Now, where am I going to my x2, well, I am again jumping to 10. Well, what would be my x3 now? So, if you have already unable to see whatever x3 is going to be, then pause for a second and try to think about this. But it is not hard to see what x3 should be, because I started with 10, I jump to 0. And again, I started with 0, I jumped to 10 and my x3 would again take me to same as x1, which will be 0. And again if I keep doing this, you should see what the problem is. So, what is the problem?

The problem seems to be that we are oscillated. We started with 10, we went to 0, we again went back to 10, we again went back to 0, and we are kind of oscillating, and that is bad. So, we, which means that the algorithm is never going to get to the actual minimum, which in this particular case of x - 5 square is 5. So, we are not, never going to get to that value, so which means that there is some issue here. And if we step back and analyze where we could have gone wrong in our reasoning, it would become clear that our reasoning was as follows.

We start with a point, then move in a direction which will take us to a different point and the hope is that the direction is a good direction. Now, in each of these cases, wherever we have started x naught, x1, x2, x3 for this particular case, our direction is correct. So, if you are in x naught, you have to move to the left, which means that you have to, well, if x naught is 10, you have to move to the left, which means that you have to add a negative value to your current value. And that is exactly what we are doing. Our - f dash of x is indeed giving us a negative value, which in this case was 10, - 10 and so we move to 0.

And again when you are at x1, you have to move to the right, which means that you have to add a positive value and which is what your - f dash of x is giving you. The only problem being that it is giving you a value which is again taking you back to x naught. So, the problem is not really with the direction. So, that is the main point.

So, the problem is not the direction of movement, but the amount you move in this direction. You should have been able to see that by now. So, you start at a point, you are moving in the right direction, but you are moving too much that you are actually missing the minimum in this

particular case, and then you are doing this back and forth between 10 and 0 for this particular case.

So, the problem is, the amount of movement seems. So, you are taking a step in the direction given by your negative derivative, but then it looks like that step is too large so that it misses the minimum. So, how can we fix this problem? So, it is the problem is, if the problem was with the direction, then we need to come up with a different function which will give us a better direction.

But the problem, thankfully, is not with the direction, it is more with the amount you have moved in that direction. So, how can we fix this? What would you do to fix this? Just pause and think about this. It is not too hard to come up with a fix for this problem and that is what we will see next.

